



Techno-economic and life cycle assessment of agrivoltaic system (AVS) designs

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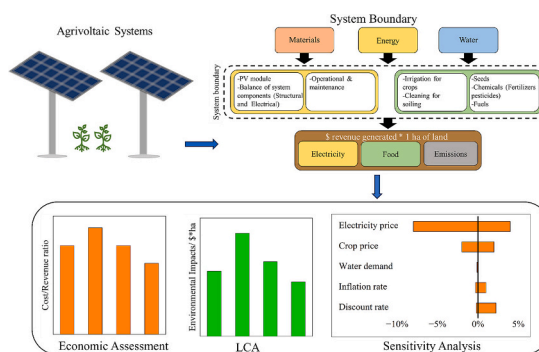
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HIGHLIGHTS

- Both life cycle economic and environmental impacts of agrivoltaics were analyzed.
- Agrivoltaics have better economic performance compared to photovoltaic-only systems.
- Agrivoltaics need 2–6¢ incentives to outperform crop-only farms.
- Agrivoltaics have 15–55 % less environmental impact than photovoltaic-only systems.

GRAPHICAL ABSTRACT



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ABSTRACT

Land use competition between agricultural activities and ground-mounted solar photovoltaic (PV) deployment has increased worldwide attention to hybrid agriculture, and PV systems known as agrivoltaic systems (AVS) in efforts to increase the efficiency of energy and food production and minimize the land use competition. However, little is known about AVS's economic feasibility and environmental tradeoffs. Here we aim to evaluate the techno-economic and environmental impacts of four AVS configurations (full density, half density, mono-axial tracking, and bi-axial tracking) and compare their performance against PV-only systems. We used the life cycle revenue generated from a hectare of land area (\$/ha) as a functional unit of our analysis. We found that all AVS configurations outperformed PV-only systems in the economic feasibility assessment, where bi-axial tracking was the best-performing AVS. Further, we developed a case scenario for agricultural farmers to determine the minimum selling price of electricity required for AVS to compete with the economic performance of crop-only farms. We found that the AVS designs require additional incentives (2¢ - 6¢ per kWh of electricity generation) to be as competitive as the crop-only farms. The life cycle environmental assessment demonstrated that the AVS has better environmental performance than PV-only systems, with ~15–55 % less environmental

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impacts per functional unit. On average, electricity generation accounts for ~80 % of AVS environmental impacts, while food production and water demand account for ~20 %. Additionally, a sensitivity analysis conducted on various uncertain parameters, such as crop yield, water demand, electricity selling price, crop selling prices, discount, and inflation rates, while varying these parameters across broader ranges, indicates that AVS designs become a more economically and environmentally sustainable alternative over PV-only systems in the majority (>66 %) of the data analyzed.

1. Introduction

The Group of Seven (G7) countries committed to decarbonizing electricity by 2035, which requires a rapid advancement of renewable energy technologies (Jordans, 2022). Solar photovoltaic (PV) electricity production is expected to play a fundamental role in reaching this goal (Silverman and Huang, 2021). In the United States (US) alone, PV electricity generation is expected to reach 419 GW by 2030 (Mackenzie, 2021). To meet these future demands, utility-scale PV systems will require 2 million acres of land, leading to land-use competition with agriculture (Dreves, 2019; Tawalbeh et al., 2021). As such, agrivoltaic systems (AVS) that generate electricity from appropriate PV designs, while allowing for simultaneous agricultural practices, have emerged as a promising solution for maintaining the synergy between food and electricity production (Agostini et al., 2021; Katsikogiannis et al., 2022; Miskin et al., 2019; Toledo and Scognamiglio, 2021; Valle et al., 2017). Global installation of AVS has been rising exponentially, from 5 MW in 2012 to 2800 MW in 2020 (Gorjian et al., 2022; I.S.E Fraunhofer, 2022).

A comprehensive systematic literature review was carried out by reviewing peer-reviewed publications and reports from non-governmental organizations that included experimental data and quantitative analyses related to AVS from 2011 to 2023. The AVS configurations of different designs of PV panels and crop types have been examined for their food and electricity production, as well as associated water demand (Al-Agele et al., 2021; Amaducci et al., 2018; Barron-Gafford et al., 2019; Marrou et al., 2013; Trommsdorff et al., 2021a; Valle et al., 2017). One of the first field applications of AVS was implemented in Montpellier, France, where the impacts of panels on crop productivity and water demand in lettuce crops were quantified (Dupraz et al., 2011; Marrou et al., 2013). The studies showed that panels with more inter-row spacing have a negligible effect on crop yield while reducing water demands by 20 % (Dupraz et al., 2011; Marrou et al., 2013). Another study on a similar application showed that an AVS incorporating solar tracking devices – panels being oriented towards the sun – improves land-use efficiency by 50 % (Valle et al., 2017). A more detailed study from Po Valley, Italy, based on a similar tracking configuration, showed that the maize yields under AVS were slightly lower when water was not a constraint, but the yields were increased when water was a constraint (Amaducci et al., 2018). Comparable results were also confirmed in the studies from Germany, where the AVS was analyzed in hybrid systems harmonizing bifacial PV panels and different crops (e.g., wheat, potato, celeriac, and grass-clover). The researchers concluded that AVS has a greater potential in arid and semi-arid climatic regions due to increased yields of wheat and potatoes observed under hot and dry summer conditions (Trommsdorff et al., 2021a; Weselek et al., 2021). Recently, the National Renewable Energy Laboratory investigated microclimatic conditions, plant drought stress, PV panel heat stress, and water use efficiency under AVS in the drylands of Arizona, US, and found that peppers and cherry tomatoes grown under AVS produced more than those of conventional farming, even when the PV panels' heat stress reduced (Barron-Gafford et al., 2019). Another study conducted in Oregon, US, demonstrated that water demand for tomato crops under AVS was less compared to full sun conditions. Furthermore, the PV concepts of innovative applications, such as vertical PV systems (Campana et al., 2021; Riaz et al., 2021) and emerging technologies (Colantoni et al., 2018; Cossu et al., 2020, 2018; Hassanien et al., 2018; Hollingsworth et al., 2020; La Notte et al., 2020;

Magadley et al., 2020; Zisis et al., 2020) (e.g., semi-transparent PVs, spherical silicon micro-PV cells, organic PVs, and dye-sensitized PVs), were also explored for their integrations through AVS concept. The findings of these studies have higher uncertainty since they are based on smaller-scale field applications (Toledo and Scognamiglio, 2021).

Most prior work on AVS was focused on assessing crop productivity, water demand, and energy efficiency. Yet, if AVS is to play an essential role in supplying electricity and food globally, it is critical to understand their environmental impacts and economic feasibility. To date, a few life cycle assessments (LCA) and techno-economic studies have been conducted on AVS (Agostini et al., 2021; Hollingsworth et al., 2020; Manoj Kumar and Chopra, 2023; Poonia et al., 2022) with only one (Agostini et al., 2021) providing an assessment of both of these indicators. However, they excluded the impact of agricultural production from the analysis. Our study aims to evaluate the economic feasibility and environmental impacts of AVS used in four distinct designs. We first conducted a systematic literature review to determine different AVS's crop productivity and water demand parameters. Second, we modeled the electricity production of each AVS design and analyzed the revenues generated. Lastly, we assessed the performance of various AVS designs to determine the most economically feasible and environmentally friendly PV configurations for AVS.

2. Methodology

2.1. Goal and scope

The goal of this study was to assess the environmental and economic impacts of four AVS designs, including static and dynamic configurations, and compare them to PV-only systems and crop-only systems. The results of this study will be useful in informing decision-makers and the public about AVS's environmental and economic benefits.

We followed the International Organization for Standardization (ISO) 14040 and 14044 (ISO 14040:2006, 2006; ISO 14044:2006, 2006) guidelines for conducting an LCA on different AVS designs. In addition, we followed methodological guidelines provided by Task 12 of the International Energy Agency (Frischknecht et al., 2020) for conducting a PV LCA. Our system boundary includes the cradle-to-farm-gate approach for crop production and the cradle-to-end-of-use phase for PV systems (Fig. 1). The PV systems' cradle-to-end-of-use phase system boundary includes raw material production, monocrystalline silicon (c-Si) PV module manufacturing, the balance of system (BOS), and operational maintenance (O&M) required for the PV systems over 30 years. The cradle-to-farm-gate approach for crop production includes irrigation, electricity, fertilizers, and pesticide demand for lettuce. We excluded end-of-life phases of both PV and crop systems because of the US's lack of federal-level regulations (El-fayome et al., 2023). The functional unit is revenue (USD) generated by AVS deployed in hectares of land (\$/ha) over their lifetime. This monetary-based functional unit encompasses the synergistic benefits and outputs derived from both agriculture and electricity production. Leon et al. also recommended using a monetary-based functional unit for conducting LCA on AVS (Leon and Ishihara, 2018). The relevant data were extracted from the Ecoinvent V.3.8 Database (Ecoinvent, 2023). Both GaBi ts 10.0 (Sphera, 2023) and OpenLCA 1.10.3 (GreenDeLTa, 2023) were used as an LCA tool. In both software, we calculated the life cycle impacts using the same impact assessment method, Tool for the Reduction and Assessment

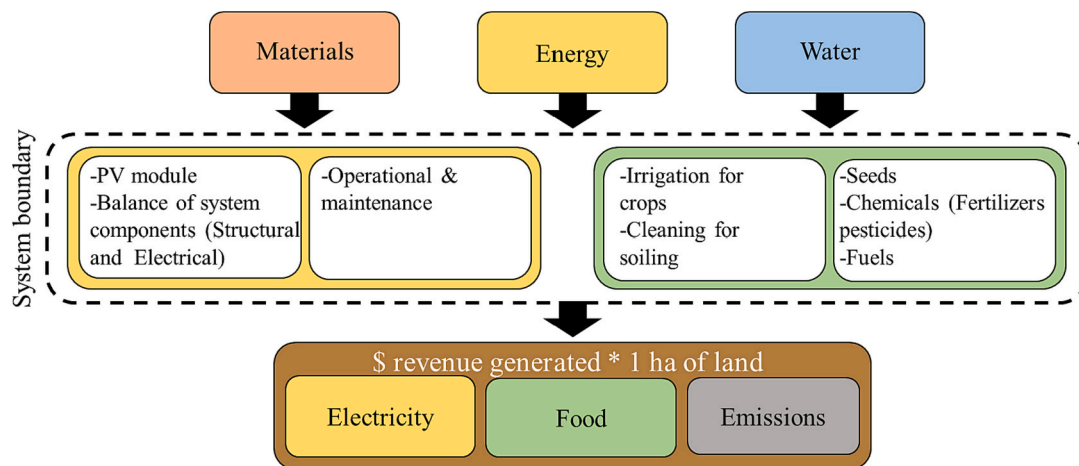


Fig. 1. System boundary for combined electricity and food production using AVS designs. We focus on the impacts of food and electricity production, and associated water demand in AVS. The revenue generated in 1 ha of land area (\$/ha) was chosen as a baseline for assessing AVS designs.

of Chemicals-TRACI 2.1 (Bare, 2012).

2.2. Modeling

We considered two static (Fig. 2a and b) full-density (FD) and half-density (HD), as well as two dynamic AVS configurations, including mono-axial (1-axis) and bi-axial (2-axis) (Fig. 2c and d). These configurations incorporate different row spacing, height, and tracking specifications. The row spacing and height of the panels were optimized and tested in the literature to prevent shading (Valle et al., 2017) and the tracking options were explored to maximize the crop yield (Amaducci et al., 2018; Valle et al., 2017). As a result, a variety of AVS types with many specifications, resulting in different crop and electricity yields from available lands, are incorporated into the analysis. For static AVS, we modeled modules mounted on stilts 4 m above the ground (Valle et al., 2017). The FD static system has a row spacing of 1.6 m, which is similar to conventional ground-mounted PV systems (Valle et al., 2017).

The HD system of the static designs has a row spacing of 3.2 m that allows more light to reach the crop (Ramos-Fuentes et al., 2023; Valle et al., 2017). The tracking systems are mounted at an elevated height to accommodate the operation of tracking systems and agriculture machinery in dynamic AVS configurations (Valle et al., 2017). In the 1-axis dynamic system, stilts were placed at 6 m spacing, and panels were mounted at 5 m with rotating capability (Valle et al., 2017). In the 2-axis dynamic system, stilts were separated by 9 m, and panels were mounted at 5 m with rotations in two orthogonal directions (Amaducci et al., 2018). The 2-axis system requires more row-to-row space than 1-axis systems due to its complex mechanism, which enables the movement of panels in both directions without collisions and prevents shading on the panels.

For the electricity yield modeling of each AVS configuration, we employed the PVSyst tool (Version 7.4) (PVSyst, 2023). Our assessment was based on 1 ha of agricultural land in Sioux Falls, South Dakota, US, one of the highest agricultural productivity locations in the state (U.S.

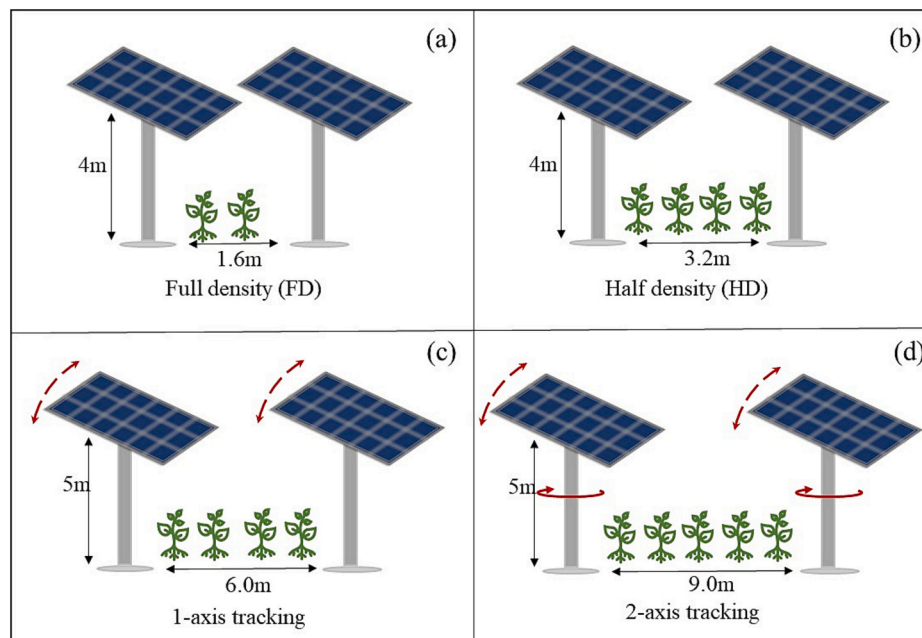


Fig. 2. Different designs of AVS, where (a) is full density (FD), (b) half density (HD), (c) mono-axial tracking (1-axis), and (d) bi-axial tracking (2-axis). Note that this study also includes a PV-only system to provide a reference point for the analysis. All the specifications of PV-only systems are identical to the FD system, except for the absence of crop production between the rows, as shown in Fig. 1a.

Table 1

Aggregated life cycle inventory for electricity production, food production, and water demand per 1 ha of land area for 30 year lifetime. Note that regarding the inverters' data, the lifetime was considered 15 years; therefore, we assumed it needed to be replaced for all the PV systems in their project lifetime (Mason et al., 2006). See Table S.2 for additional specifications for the AVS designs.

		Component	Units	Crop-only	Static systems		Dynamic systems		References
					HD	FD	1-Axis	2-Axis	
Electricity	PV	c-Si PV module	m ²	–	3033	6070	3239	2158	(Frischknecht et al., 2020)
		Electricity for tracking	kWh	–	–	–	41,038	27,342	(Sinha et al., 2013)
	Structural BOS	Steel	Tonnes	–	73	152	137	138	(Agostini et al., 2021; Sinha and de Wild-Scholten, 2012)
		Aluminum	kg	–	520	1042	606	432	(Sinha and de Wild-Scholten, 2012)
		Zinc coat	kg	–	573	1147	725	483	(Sinha and de Wild-Scholten, 2012)
		Concrete	kg	–	11,353	22,720	10,500	8167	(Agostini et al., 2021; Sinha and de Wild-Scholten, 2012)
	Electrical BOS + inverter	Copper	kg	–	2640	5280	3777	2590	(Sinha and de Wild-Scholten, 2012)
		Polyethylene composite	kg	–	850	1700	1360	1080	(Sinha and de Wild-Scholten, 2012)
		Tracking motor	kg	–	–	–	476	377	(Agostini et al., 2021)
		Transformer	kg	–	5349	10,697	7206	5605	(Mason et al., 2006)
		Inverters (30 kW)	–	–	18	33	21	19	(PVsyst, 2023)
		Crop field	m ²	8000	6967	3930	6761	7842	
Food	Lettuce production	Romaine seed	Million	46.6	32.4	18.2	31.4	36.5	(Nicholson et al., 2020)
		N fertilizers (as N)	Tonnes	9.00	6.24	3.48	6.00	6.96	
		P fertilizers (as P ₂ O ₅)	Tonnes	9.60	6.60	3.72	6.48	7.44	
		K fertilizers (as K ₂ O)	Tonnes	24.0	16.68	9.48	16.20	18.72	
		Fungicides	Tonnes	0.24	0.17	0.10	0.16	0.19	
		Herbicides	Tonnes	0.48	0.34	0.19	0.32	0.37	(Bartzas et al., 2015; Romero-Gómez et al., 2014)
		Insecticides	Tonnes	0.36	0.24	0.14	0.24	0.28	
		Diesel for machinery	Gallon	1404	984	552	948	1104	(Romero-Gómez et al., 2014)
	Crop	Water for irrigation	M m ³	0.43	0.21	0.08	0.20	0.28	(Nicholson et al., 2020)
		Electricity for irrigation	MWh	255	125	50	121	170	(Sowby and Dicaldo, 2022)
Water	PV	Water for PV cleaning	m ³	–	180	360	192	132	(Jones et al., 2016)
		Energy for PV cleaning	MWh	–	0.11	0.22	0.12	0.07	(Sowby and Dicaldo, 2022)

Department of Agriculture, 2019). The average annual local solar irradiation at this location is 1840 kWh/m² (Sengupta et al., 2019). The solar module selected for this simulation was Trina Solar TSM-DE14A(II) 380Wp (1.96 m × 0.92 m) which is a crystalline silicon (c-Si) type solar PV module (TrinaSolar, 2023). In static systems, panels were tilted at an optimum angle of 30° for maximum electricity output (Majumdar and Pasqualetti, 2018). In 1-axis tracking systems, panels were oriented north to south and tracked the sun from –60° to +60° in the east-west direction, whereas 2-axis panels moved from east to west and north to south (Agostini et al., 2021). Additional technical specifications of PV-only and different AVS designs can be found in Table S.1.

Leafy vegetables such as spinach, escarole, and lettuce are common in AVS studies since they are shade-tolerant (Elamri et al., 2018; Jung et al., 2021; Valle et al., 2017). Therefore, we selected Romaine lettuce production for this study and used a rate of 2.0 kg lettuce per 1 m² as a baseline for crop yield modeling (Tourte et al., 2019). Crop production for each AVS design was calculated, considering both the shading impact caused by PV panels and the available active land area (i.e., ground not covered by solar panels). The active land area available for FD, HD, 1-axis, and 2-axis designs was 50 %, 12 %, 15 %, and 2 % less than farms without PV panels, respectively. Based on the studies (Elamri et al., 2018; Valle et al., 2017), we determined the mean crop yield reduction values due to shading as 40 %, 20 %, 12 %, and 5 % for FD, HD, 1-axis, and 2-axis systems, respectively.

2.3. Life cycle inventories

Table 1 summarizes specific components required for electricity production, food production, and water demand in the four AVS configurations for the 1 ha implementation area. Primary and secondary data were used to prepare inventories required for food, electricity

production, and water demand in each of the four AVS configurations. We modeled four lettuce growth cycles in a year per ha cropland (Jung et al., 2021). For water demand, we considered the impact of reduced irrigation due to shading around 30 % in HD, 50 % in FD, 30 % in 1-axis, and 15 % in 2-axis AVS designs (Elamri et al., 2018; Ramos-Fuentes et al., 2023). Additionally, for PV panel water requirements, we assumed four cleaning cycles per year during the O&M phase (Jones et al., 2016).

2.4. Economic analysis

The cost inventories for electricity production, food production, and water demand are shown in Table 2. The electricity production cost inventories include the cost of PV modules, structural and electrical BOS, inverters, and O&M required for the PV systems. According to US National Renewable Energy Laboratories reports, the annual O&M cost for utility-scale systems is \$16 per kW-year (Ramamany et al., 2022). We excluded inverter replacement costs from the PV system's annual O&M calculations and used \$13 per kW-year as the annual O&M cost. The inverters need to be replaced after 15 years; therefore, we calculated the present worth of inverter replacement cost separately. The cost inventories for herbicides, fungicides, and insecticides were calculated by taking the prices of commonly used products in lettuce crop production (Agricultural Chemical Solutions, 2023; Lawn and Pest Control Supply, 2023; Unglesbee, 2022).

The economic analysis of various AVS designs was carried out by calculating the life cycle cost, life cycle revenue, and cost/revenue ratio values. The cost/revenue ratio is the ratio of life cycle costs to life cycle revenue expressed in present worth (Eq. (1)). A lower cost/revenue ratio value indicates greater economic viability of an AVS design.

Table 2

Cost inventories for electricity, crops, and water demand. * Indicates the electrical BOS cost for static AVS, while ** indicates the electrical BOS cost for dynamic AVS. The electrical BOS cost of dynamic configurations includes the cost of electrical motors. The annual O&M cost includes PV module cleaning, module and components replacement, and other costs.

	Component	Price/unit	Unit	Reference
Electricity	c-Si module	0.40	W _{DC}	(Ramasamy et al., 2022)
	Steel	0.87	kg	(SteelBenchmarker, 2023)
	Aluminum	2.52	kg	(Statista, 2023)
	Zinc	3.22	kg	(U.S. Geological Survey, 2022)
	Inverters	0.08	W _{DC}	(Horowitz et al., 2020)
	Concrete	0.07	kg	(Ayoola Azzan, 2023)
	Electrical BOS	0.27*	W _{DC}	(Horowitz et al., 2020)
		0.30**	W _{DC}	
Food	O&M	13	kW-yr	(Ramasamy et al., 2022)
	Seed	1.40	Thousand	(Nicholson et al., 2020)
	N fertilizers (as N)	1.83	kg	(Plastina, 2023)
	P fertilizers (as P ₂ O ₅)	1.65	kg	
	K fertilizers (as K ₂ O)	1.58	kg	
	Herbicides	44	kg	(Unglesbee, 2022)
	Fungicides	50	kg	(Lawn and Pest Control Supply, 2023)
	Insecticides	17	kg	(Agricultural Chemical Solutions, 2023)
	Diesel for machinery	4.13	Gallon	(U.S. Energy Information Administration, 2023a)
Water	Water for irrigation	0.17	m ³	(Nicholson et al., 2020)
	Electricity for irrigation	0.07	kWh	(U.S. Energy Information Administration, 2023b)

$$\frac{\text{Cost}}{\text{Revenue}} \text{Ratio} = \frac{\text{Life cycle cost}}{\text{Life cycle revenue}} \quad (1)$$

The life cycle cost is the total cost of AVS over their lifetime expressed in the present instant of time by accounting for the time value of money (Braham, 2017). Life cycle cost is the sum of the cost of electricity production, food production, and irrigation (Eq. (2)).

$$\text{Life cycle cost} = C_{\text{Electricity}} + \sum_{n=1}^{n=30} C_{\text{Food}} (1+r)^n \frac{(1+i)^n - 1}{i(1+i)^n} + \sum_{n=1}^{n=30} C_{\text{Irrigation}} (1+r)^n \frac{(1+i)^n - 1}{i(1+i)^n} \quad (2)$$

where $C_{\text{Electricity}}$ is the total electricity production cost, C_{Food} is the annual cost of food production in year n , $C_{\text{Irrigation}}$ is the annual cost of irrigation in year n , r is the inflation rate, and i is the discount rate used to calculate the present worth of future costs. $C_{\text{Electricity}}$ was calculated as:

$$C_{\text{Electricity}} = C_{\text{PV}} + C_{\text{BOS}} + C_{\text{Inverter}} + \sum_{n=1}^{n=30} C_{\text{O\&M}} (1+r)^n \frac{(1+i)^n - 1}{i(1+i)^n} \quad (3)$$

where C_{PV} is the cost of the c-Si PV module, C_{BOS} is the total cost of structural and electrical BOS, C_{Inverter} is the cost of the inverter, $C_{\text{O\&M}}$ is the annual O&M cost in year n , r is the inflation rate, and i is the discount rate used to determine the present worth of future annual O&M.

Life cycle revenue is the sum of the total revenue generated from electricity production and crop production in the project lifetime expressed in present worth (Poonia et al., 2022). Life cycle benefit was calculated using Eq. (3).

$$\text{Life cycle revenue} = \sum_{n=1}^{n=30} R_{\text{Crop}} (1+r)^n \frac{(1+i)^n - 1}{i(1+i)^n} + \sum_{n=1}^{n=30} R_{\text{Electricity}} (1+r)^n \frac{(1+i)^n - 1}{i(1+i)^n} \quad (4)$$

where R_{Crop} is crop revenue generated in year n , $R_{\text{Electricity}}$ is electricity revenue generated from AVS in year n , r is the inflation rate, and i is the discount rate used to calculate the present worth of future revenues from crop and electricity. The revenue generated from electricity varies as a

function of the annual electricity yield degradation rate(d). $R_{\text{Electricity}}$ was calculated as:

$$R_{\text{Electricity}} = E_A \cdot T \times \sum_{n=1}^{n=30} (1-d)^n \quad (5)$$

where E_A is annual electricity generation, T is the selling price of electricity, and d is the annual degradation of electricity generation (0.5 %).

We used a 2.5 % inflation rate and a 5 % discount rate for this study. A 30-year lifetime was considered for all the assessed PV systems in this study. To calculate the revenue generated by electricity and crop in each design, we multiplied the quantities produced with an average selling price of electricity at 7¢/kWh (U.S. Energy Information Administration, 2023b) and lettuce at 75¢/kg (U.S. Department of Agriculture, 2023).

2.5. Uncertainty and limitations

The modeling of both techno-economic and environmental impact assessment of AVS designs included several assumptions and approximations. For example, the nominal discount rates for electricity and agricultural production activities may change over time depending on the inflation in the economy (Bankrate, 2023; Ramasamy et al., 2022). In this study, a representative federal discount rate for the US economy in 2023 was used (5 %) for both PV and agricultural systems (Bankrate,

2023). We performed a sensitivity analysis for this parameter to cover a wide range between 0.5 % and 12 % since the rates fluctuated significantly, recently, i.e., today's federal discount rates are more than twice of a year ago (Bankrate, 2023). In addition, the annual inflation rates in

Table 3

The economic performance and yield parameters of single production and hybrid production AVS designs. Additional breakdowns of cost parameters for both single and hybrid systems are presented in Table S.2. *Note that a lower cost/revenue ratio indicates that the system has better economic performance.

		Units	Single systems		Hybrid production AVS designs			
			PV-only	Crop-only	Static		Dynamic	
					HD	FD	1-Axis	2-Axis
Cost	Life cycle cost	M\$/ha	9.70	4.22	7.80	11.40	9.37	8.38
	C _{Electricity}		9.70		4.85	9.70	6.52	5.08
	C _{Food}			2.75	1.91	1.08	1.86	2.15
	C _{Irrigation}			1.47	1.03	0.58	1.00	1.15
Yield	Electricity	GWh/ha	46.4		24.7	46.4	35.56	32.0
	Crop	kt/ha		3.12	2.17	0.92	2.32	2.91
Revenue	Life cycle revenue	M\$/ha	50.4	37.3	52.9	61.4	66.39	69.5
	R _{Electricity}		50.4		26.9	50.4	38.66	34.7
	R _{Crop}			37.3	26.0	11.0	27.73	34.7
	Cost/revenue ratio*		0.19	0.11	0.15	0.18	0.14	0.12

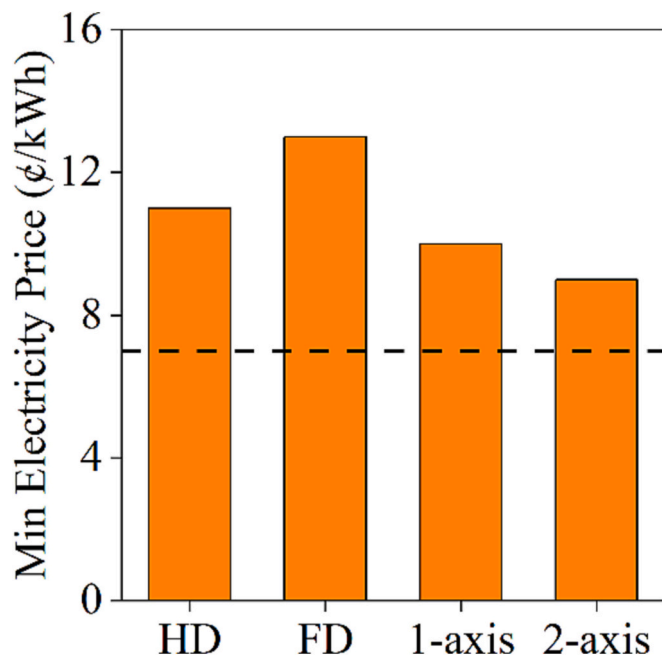


Fig. 3. The minimum electricity selling price that agricultural farmers should receive per respective AVS design to meet a crop-only farm's cost/revenue ratio. The dashed line indicates the average selling price (7¢/kWh) considered in this study.

the US over the past decade exhibited significant fluctuations, ranging from as low as 0.7 % to as high as 6.5 % (CoinNews Media Group, 2023). Thus, our assessment also considered a broad range of inflation rates, spanning from 0.5 % to 6 %. Next, the electricity selling and crop prices in the US vary in different jurisdictions depending on the available AVS incentives (Dawnbreaker, 2022). Therefore, we also analyzed the impact of electricity selling prices, which varied from a minimum of 2¢ per kWh to the average selling price of residential electricity in the US, which is 12¢ per kWh (U.S. Energy Information Administration, 2023b). We considered this wide range of electricity selling prices to provide a more comprehensive assessment of the impact of local incentives within the states. Incorporating crop yield reductions due to shading and water demand from other studies can introduce a significant level of uncertainty, which may vary depending on the geographic location (Sarr et al., 2023). The selling price of lettuce also varied depending on the geographic location. Therefore, for crop selling price, crop yield, and water demand, we conducted sensitivity analysis by adjusting them within the range of −30 % to +30 % relative to values considered in this study. A typical range that's used in uncertainty analysis of AVS-specific reports of the US National Renewable Energy Laboratories is ±25 % (Horowitz et al., 2020) and we selected a slightly broader range to provide a more comprehensive sensitivity analysis. Other parameters that may have uncertainty are row distance, mounting height, and tracking type. However, these factors were excluded from the sensitivity analysis as their impact was already captured in the four AVS scenarios analyzed in this study. In addition, we lack experimental data demonstrating a direct relationship between these parameters and crop yield, water demand, and consequently, revenue from AVS. An important note to consider is that our study focused on the scenario where farmers own the land and operate AVS themselves, which is a representative case for

Table 4

Normalized environmental impacts of single production and hybrid production AVS per \$ revenue generation from a unit land area (1 ha). The results are normalized concerning the environmental impacts of the PV-only system (last column). Normalization and averaging of the impacts are done similarly to the approach followed by Celik et al. (2017). This approach serves to enhance result interpretability and facilitate decision making. All the LCA raw data regarding hybrid and single systems can be extracted by using the representative multipliers and baseline data of the PV-only system, provided in the last column.

		Hybrid production AVS designs				Single systems		PV-only (/ \$ per ha)
		HD	FD	1-Axis	2-Axis	Crop-only	PV-only	
Impact categories	Acidification	0.56	0.87	0.54	0.43	0.20	1.00	3.90E-03 (kg SO _{2-eq})
	Ecotoxicity	0.47	0.84	0.51	0.38	0.04	1.00	4.84E+01 (CTU _e)
	Eutrophication	0.70	0.94	0.70	0.60	0.15	1.00	4.08E-03 (kg N _{eq})
	Global warming	0.60	0.88	0.62	0.54	0.33	1.00	4.03E-01 (kg CO _{2-eq})
	Human health particulate	0.43	0.84	0.51	0.39	0.002	1.00	1.01E-01 (kg PM _{2.5-eq})
	Human tox., cancer	0.43	0.84	0.51	0.39	0.00	1.00	2.73E-04 (CTU _h)
	Human tox., non-can	0.58	0.87	0.58	0.46	0.21	1.00	5.42E-07 (CTU _h)
	Ozone depletion	0.46	0.85	0.52	0.41	0.06	1.00	3.12E-07 (kg CFC 11 _{eq})
	Resources	0.77	0.94	0.71	0.65	0.70	1.00	2.71E-01 (MJ energy)
	Smog air	0.46	0.84	0.52	0.40	0.02	1.00	1.20E-01 (kg O _{3-eq})
	Average	0.55	0.87	0.57	0.46	0.18	1.00	



Fig. 4. The breakdown of environmental impacts of PV panels, structural and electrical BOS, food production, and water demands in each impact category across all AVS designs.

the majority of the AVS sites operating in the US (Casey, 2023; Dreves, 2022). Other scenarios may include leasing the land to a solar investor or engaging in co-ownership of AVS (Macknick et al., 2022; Trommsdorff et al., 2021b), which were not explored in this study due to a lack of data on financial parameters such as cost premiums, insurance rates, and lease payments.

3. Results and discussion

3.1. Economic assessment

All AVS designs outperformed PV-only systems, and the 2-axis system is the best-performing AVS design in terms of cost/revenue ratio (Table 3). The AVS designs have a lower cost/revenue ratio due to their higher life cycle revenue when compared to PV-only systems. The life

cycle revenue of the AVS is higher (5–40 %) than that of a PV system, while the life cycle cost is 3–20 % lower (except for FD). The FD system has nearly 15 % more life cycle costs than PV-only systems. The high economic performance of the 2-axis system among AVS designs is due to two primary reasons: First, crop revenue from the 2-axis systems is similar to crop-only systems because the impact of shading is minimal (5 % less compared to crop-only systems). Second, the life cycle cost of 2-axis systems is lower compared to FD and 1-axis systems since fewer PV systems can be located per hectare of land.

To further assess the AVS design's economic feasibility, we developed a case scenario to identify the incentives required for agricultural farmers to keep their current crop-only cost/revenue ratio (i.e., 0.11). Fig. 3 shows the minimum selling price of electricity that a farmer should receive for each AVS design to meet the economic performance of a crop-only farm. Our results show that to compete with the cost/

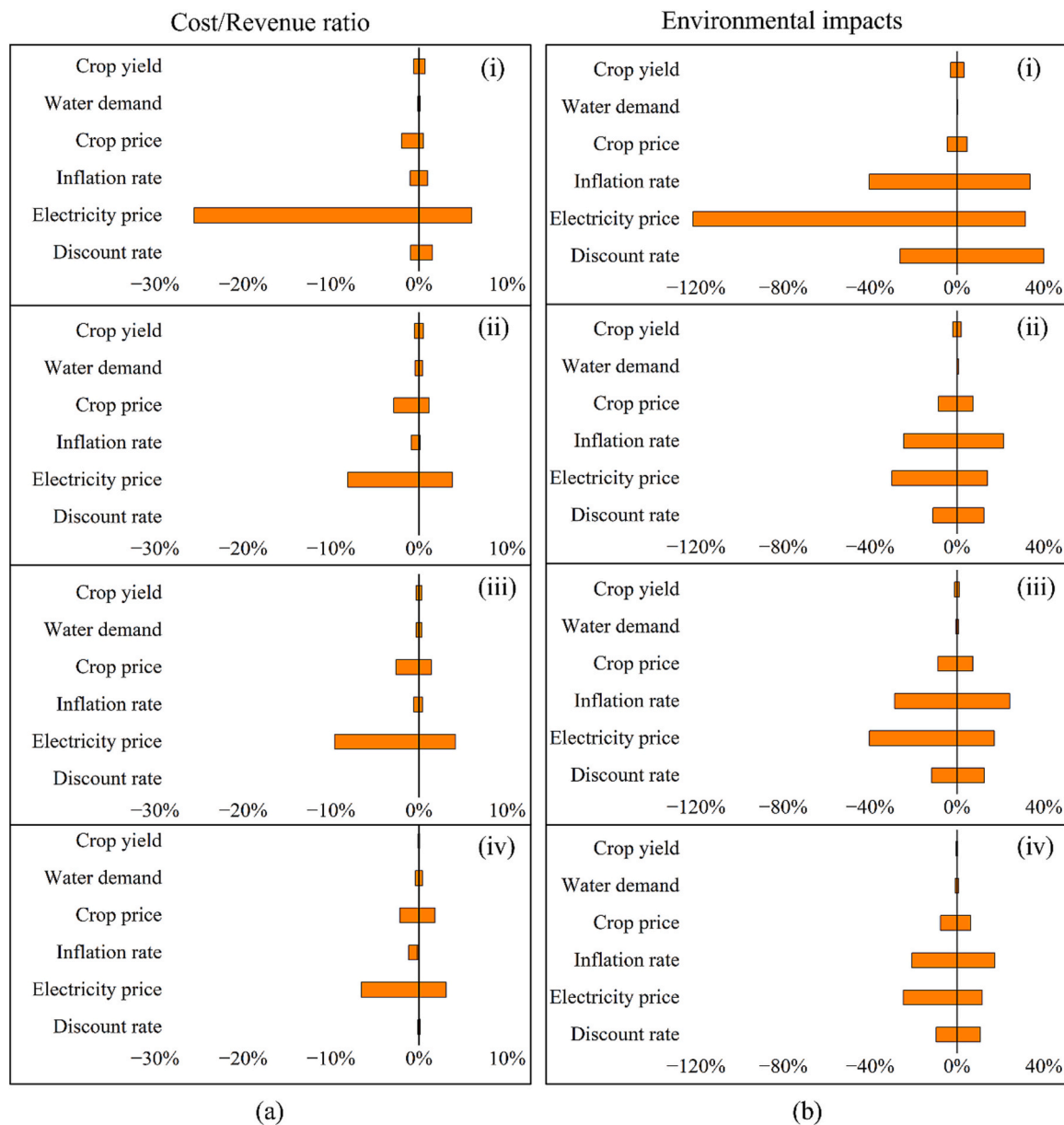


Fig. 5. The sensitivity of cost/revenue ratio (a) and the average normalized environmental impacts (b) of FD (i), HD (ii), 1-axis (iii), and 2-axis (iv) AVS designs due to change in crop yield, water demand, crop selling price, inflation rate, electricity selling price, and discount rate.

revenue ratio of the crop-only farm, the HD, FD, 1-axis, and 2-axis systems would require additional incentives of 4¢/kWh, 6¢/kWh, 3¢/kWh, and 2¢/kWh, respectively. These values are within the suggested incentive levels proposed in the US. For example, the Solar Massachusetts Renewable Target program supported by the US Department of Energy offers 6¢/kWh incentives for farmers to deploy AVS designs in addition to the base compensation rate (Dawnbreaker, 2022).

We also compared the results with the relevant literature. For example, Dinesh et al. found that HD and FD systems can produce 8 % and 30 % more revenue than conventional lettuce production, respectively (Dinesh and Pearce, 2016). Similarly, our study revealed comparable outcomes: HD and FD systems yielded 5 % and 20 % more revenue, respectively, compared to crop-only farms. In a separate study on economic analysis, a 2-axis AVS generated higher revenue than conventional PV over the project's lifetime (Agostini et al., 2021). These results are consistent with our results. However, we note that Agostini et al. (2021) excluded the economic benefits of agricultural activities. Another economic analysis conducted on an HD-type AVS design

revealed that the life cycle revenue of HD was 1.5 times its life cycle cost (Poonia et al., 2022), whereas we found HD had ~7 times its life cycle cost. The higher life cycle revenues in our study are primarily attributed to the lower discount rates (5 %), whereas Poonia et al. (2022) assumed 12 % in their study.

3.2. Environmental impact assessment

The environmental impacts of different AVS and single systems are shown in Table 4. The PV-only system was chosen as the reference point for this comparative analysis, and each design's environmental impact from each impact category was normalized according to the PV-only system. Overall, all four AVS designs outperformed the PV-only system in terms of environmental performance. The average normalized environmental impacts per \$1 of revenue generation per 1 ha of land of AVS are ~15–55 % less than PV-only systems. Among AVS designs, the 2-axis system demonstrated the best performance, whereas the systems that utilized densely deployed PV systems (i.e., FD) resulted in more

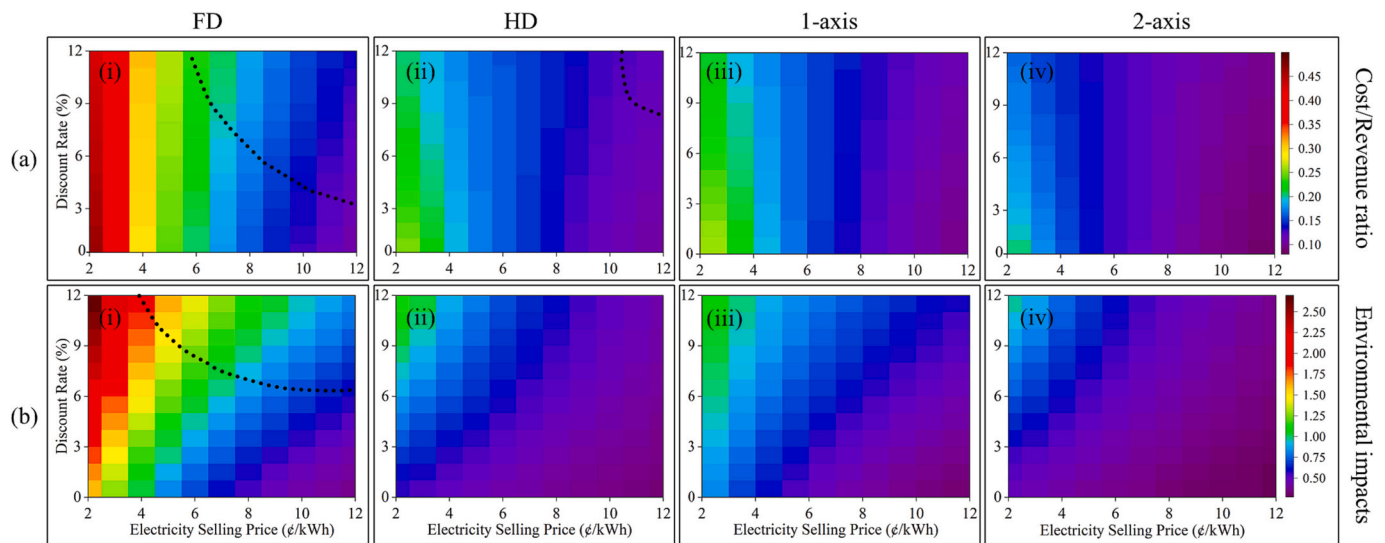


Fig. 6. The impact of discount rate and the electricity selling price on the cost/revenue ratio (a) and the average normalized environmental impacts (b) of FD (i), HD (ii), 1-axis (iii), and 2-axis (iv) AVS designs, respectively. The left portion of the dotted lines in each graph shows the values where the cost/revenue ratio and the averaged environmental impacts of the AVS outperform PV-only systems. The figures without dotted lines indicate that AVS designs outperform PV-only systems at all discount rates and electricity prices. Additional sensitivity analyses of other uncertainty parameters are provided in Figs. S1 and S2.

environmental impacts across all impact categories. The superior environmental performance of 2-axis systems is primarily attributed to their higher revenue generation (15–50 % more) compared to other AVS designs. The increased environmental impacts of the FD system arise from both the greater number of PV panels (1500–2000 more per ha) and the corresponding BOS requirement per unit of land area. The HD and 1-axis systems have ~45 and 40 % fewer impacts, respectively, compared to a PV-only system, while the crop-only farm has ~80 % fewer impacts than PV-only systems.

On average, ~80 %, 10 %, and 10 % of the impacts in each category were attributed to electricity production, food production, and water demand, respectively (Fig. 4). The higher impacts due to electricity production of each AVS design in each impact category were mainly attributed to structural BOS (20–50 %), electrical BOS (5–30 %), inverters (10–20 %), and PV panels (10–40 %). The structural BOS components, such as steel and zinc coating, are primarily responsible for the majority of the energy-production impacts in each category. Because of the higher mounting heights, AVS designs require a greater amount of steel and zinc coating, resulting in more impacts from these components in each category. These results are consistent with the literature; Agostini et al. (2021) also reported that BOS components contribute 50–60 % of the 2-axis AVS design's global warming potential impacts (Agostini et al., 2021). The researchers also underlined that structural BOS has a significant impact (Agostini et al., 2021). The inverters used in AVS are responsible for ~50 % of the ecotoxicity impacts. The higher impacts of inverters in this category are primarily due to the use of integrated circuits, printing wiring boards, and copper. Electrical BOS, such as copper and polyethylene and transformers, contributed 30–40 % of eutrophication and human toxicity cancer impacts. PV panels contribute significantly to global warming, human health particulate air, resource, and ozone depletion categories, accounting for 30–50 % of total AVS impacts. The impacts of PV panel production were dominated by solar-grade silicon production, as reported in previous life cycle assessment studies on PV systems (Celik, 2018; Celik et al., 2020; Fthenakis and Leccisi, 2021).

The food production impacts contributed up to 20 % of resources, smog, and ozone depletion categories within each AVS. Fertilizers were primarily responsible for food production impacts in these categories (40–65 %). The water demand impacts accounted for water used for crop irrigation and water used for PV panel cleaning during operation. Water demand impacts contributed 15 % to resource and ozone

depletion categories and have small impacts in the remaining categories. Outside this study, only a few studies conducted environmental analysis on crop production under AVS designs (Hollingsworth et al., 2020; Leon and Ishihara, 2018). These studies incorporate a greenhouse system for crop production and found that the environmental impacts of tomato production under the AVS were less compared to conventional greenhouse systems. These results are contradictory to our results because crop production parameters in greenhouses differ significantly from those in the open-field cultivation approach in our study.

3.3. Sensitivity analysis

Fig. 5a and b was prepared to show the impact of the variability in six key parameters (e.g., crop yield, water demand, crop price, inflation rate, electricity selling price, and discount rate) on the cost/revenue ratio and average normalized environmental impacts of four AVS designs, respectively. Fig. 5a indicates that the electricity selling price is the most sensitive parameter on the cost/revenue ratio, while minor impacts were also noticed in the discount rate, inflation rate and crop price. The impact of the varied electricity price affects the AVS designs by up to ~30 %. We also noted that the impact of the variability in the inflation rate, electricity selling price, and discount rate of the FD system is more than the other AVS designs since the contribution of $R_{\text{Electricity}}$ to the life cycle revenue of the FD system (~80 %) is higher compared to the other AVS design. On the contrary, cost/revenue ratios of HD, 1-axis, and 2-axis systems have slightly higher sensitivities due to the variability in crop price, as these systems have higher impacts on the life cycle revenue contributed by R_{Crop} (up to ~50 %).

Moreover, the electricity selling price is also the most sensitive parameter in the calculation of the average normalized environmental impact of AVS designs (Fig. 5b). However, both the inflation rate and discount rates have a moderate influence as well (~10–40 %). Of both cost/revenue ratio and environmental impact, the least sensitive parameters are crop yield reductions due to shading and water demand for the crop. We also found variations in these sensitivity parameters lead to greater uncertainty in environmental impacts compared to the cost/revenue ratio. This is because the variability in the assessed parameters (i.e., particularly the discount and inflation as seen in Eq. (2)) affect both components of cost/revenue ratio calculations (as shown in Eq. (1)); thus, their impacts are effectively reduced. However, the environmental impact assessment (e.g., total life cycle impacts/life cycle revenue) is

only influenced by the changes in life cycle revenue (i.e., while the total life cycle impacts are unchanged) which resulted in higher sensitivity of the assessment depending on uncertain parameters.

The discount rate and the selling price of electricity are the two most influential parameters identified in Fig. 5 and the impact of these factors on the cost/revenue ratio and the averaged environmental impacts are shown in Fig. 6a and b, respectively. Fig. 6a shows that the cost/revenue ratio of FD, HD, 1-axis, and 2-axis systems ranges from 0.08 to 0.5, with lower cost/revenue values observed in the 2-axis system. In addition, we showed the region where AVS designs produce a lower cost/revenue ratio than PV-only systems at specific electric selling prices and discount rates (the left portion of the dotted line). As shown in Fig. 6a, in most of the cases analyzed (>66 %), AVS designs outperform PV-only systems. The average normalized environmental impacts of FD, HD, 1-axis, and 2-axis systems vary from 0.27 to 2.65 (Fig. 6b), with lower environmental impacts observed in 2-axis systems as a function of discount rate and electricity selling price. Except for FD systems, our results show that the remaining AVS designs outperform PV-only systems at all discount rates and electricity selling prices, while 68 % of FD scenarios outperform PV-only systems. Additional analysis exploring the impact of the inflation rate and crop selling price on the cost/revenue ratio and averaged normalized environmental impacts are also provided in Figs. S.1 and S.2.

4. Conclusions

We analyzed the four AVS designs to estimate their economic and environmental sustainability compared to PV-only and crop-only farms. Our economic analysis demonstrates that all AVS designs outperformed PV-only systems and the system that utilized PV panels that rotated on the 2-axis was the best-performing AVS design in terms of the cost/revenue ratio. Further, the case scenario developed for farmers demonstrates that HD, FD, 1-axis, and 2-axis systems require additional incentives of 4¢/kWh, 6¢/kWh, 4¢/kWh, and 2¢/kWh, respectively to compete with the crop-only systems. Additionally, all AVS designs had better environmental performance than PV-only systems. The AVS designs had ~15–55 % lower environmental impacts per \$ revenue generation per 1 ha of land than PV-only systems. We found that 80 % of the environmental impacts were attributed to electricity generation, while food production and water demand accounted for only 20 %. The sensitivity analysis performed on various uncertainty parameters, broadening the range of variation, demonstrates that AVS designs have better economic and environmental performance over PV-only systems in majority of the analyzed data (>66 %).

CRediT authorship contribution statement

Achyuth Ravilla: Writing – original draft, Visualization, Software, Methodology, Data curation, Conceptualization. **Gabriela Shirkey:** Writing – review & editing. **Jiquan Chen:** Writing – review & editing, Validation. **Meghann Jarchow:** Writing – review & editing, Validation. **Olyssa Stary:** Writing – review & editing. **Ilke Celik:** Writing – review & editing, Visualization, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2023.169274>.

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